Software Engineering

Unit-wise Previous Year Questions

Chaitanya Bharathi Institute of Technology — compiled from 13 papers (Dec 2021 – Nov 2025), mapped to the R22 syllabus. Questions outside the syllabus are left out; repeated questions are merged.

How to read	
3×	Asked in 3 different papers — highest priority.
Short	Part-A style (2–3 marks).
Long	Part-B style (4–10 marks), grouped by syllabus topic.

Unit	Short	Long	Repeated
UNIT I — Introduction & Process Models	11	27	8
UNIT II — Agile Development & Requirements	9	29	6
UNIT III — Project Planning & Estimation	9	31	11
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UNIT V — Testing, Reviews, Quality & Deployment	13	31	20

Source papers

Nov'25	22CSC21 — V Sem (CSE, IT, AI&DS), Nov 2025 — R22
May'25	22CSE21 — VI Sem (CET), May 2025 — R22
Dec'24	22CSC21 — V Sem (CSE/IT/AI&DS), Dec 2024 — R22
Jun'24	20CSC22 (CET) & 20ITC13 (AI&DS) — May–June 2024 — R20
Dec'23	20CSC22 — V Sem (CSE, CSM), Dec 2023 — R20
Aug'23	20ITC13 — IV Sem (IT), Aug 2023 — R20
Jun'23	20CSC22 (CSE-IoT) & 20ITC13 (AI&DS) — May/June 2023 — R20
Jan'23	20CSC22 — V Sem (CSE, CSE-AI&ML), Dec 2022/Jan 2023 — R20
Aug'22	20ITC13 — IV Sem (IT), July/Aug 2022 — R20
Jun'22	18CSC24 — VI Sem (CSE), June 2022 — R18
Dec'21	18ITC18 — V Sem (IT), Dec 2021 — R18

UNIT I — Introduction & Process Models

SE practice, software process, generic process model, process assessment & improvement, Waterfall, Incremental, RAD, Prototyping, Spiral, Specialized process models

Short Answer Questions

1.	Define Software and Software Engineering. List the characteristics of software engineering.	2× Jun'23, Jun'24
2.	Define the term Software Engineering Practice.	May'25
3.	Define software process. List its activities.	Jun'22
4.	List out the five generic process framework activities for software engineering.	Dec'21
5.	List out the umbrella activities of a software process.	Jun'22
6.	List the different approaches for software process assessment.	Dec'23
7.	Suggest two advantages and two disadvantages of the approach to process maturity embodied in the SEI Capability Maturity Framework.	Nov'25
8.	Name the successive phases in the waterfall model.	Aug'23
9.	What are the drawbacks of the Rapid Application Development (RAD) model?	Jan'23
10.	What are the key outputs during an iteration in a project following an iterative development model?	Dec'24
11.	Summarize the main differences between evolutionary software development and traditional software development.	Jun'24

Long Answer Questions

Software, SE Practice & Process Framework

1.	Describe the layered technology of software engineering / Demonstrate SE layers with a neat diagram.	2× Dec'21, Dec'23
2.	Explain the Software Process Framework.	Dec'23
3.	Umbrella activities occur throughout the software process. Are they applied evenly across the process or concentrated in one or more framework activities?	2× Dec'24, Nov'25
4.	Discuss the importance of various umbrella activities for successful project development.	Jun'23
5.	What are the Management Myths? State the reality that accompanies the Practitioner's myth.	2× Dec'24, Nov'25
6.	What is the essence of software practice? Discuss the general principles of software practice.	Jan'23
7.	“Software doesn't wear out” — explain in detail.	Jun'22
8.	What are the fundamental characteristics that define software and distinguish it from other engineering disciplines?	Jun'24
9.	Is software a product or a process? Which has more relevance? Justify with a suitable example.	Aug'22
10.	What are Software Engineering goals and approaches?	Jun'23

Process Assessment & Improvement

11.	Discuss process assessment and improvement approaches.	May'25
12.	Explain how Capability Maturity Model Integration (CMMI) is utilized in assessing software processes.	Jun'24
13.	Compare the various CMMI maturity levels for software process improvement.	Aug'22

Waterfall Model

14.	Explain the Waterfall model (with a diagram) and its advantages / What is the purpose of the waterfall model?	**5×** Jun'23, Dec'23, Jun'24, May'25, Aug'22
15.	Explain the waterfall model and analyze its limitations.	Dec'21
16.	Define Software Engineering. Why do real projects rarely use the waterfall model? Explain.	Jan'23
17.	Illustrate the Waterfall model. Provide three specific examples of software projects that would be amenable to the waterfall model.	Dec'24

Incremental & RAD Models

18.	Explain the incremental process model and its advantages.	Jun'22
19.	What is the RAD Model?	**2×** Jun'23, Jun'24
20.	Compare the RAD Model with the Prototyping Model.	Nov'25

Evolutionary Models — Prototyping & Spiral

21.	Explain / construct the Spiral model with a diagram; describe its phases and the activities in each phase.	**5×** Jun'22, Jun'23, Dec'23, Aug'23, May'25
22.	Explain why the spiral model is more useful than the prototype model in development of a software product.	Jan'23
23.	As you move outward along the spiral process flow, what can you say about the software being developed or maintained?	Nov'25
24.	Compare and contrast various software process models, including the Waterfall and Spiral models.	Jun'24
25.	Give your reasoning in choosing a suitable process model for: (i) client has sufficient budget, vendor has sufficient manpower, both know the requirements; (ii) client is cautious and wants risk assessment at each stage.	Jun'23

Specialized Process Models

26.	What are specialized process models? Explain any two / Compare various specialized process models.	**2×** May'25, Jan'23
27.	Describe the component-based and formal methods models for software development, with their advantages and disadvantages.	Dec'24

UNIT II — Agile Development & Requirements

Agility, XP, ASD, Scrum, DSDM, FDD, Agile Modelling; Requirements engineering — groundwork, elicitation, negotiation, validation; SRS — value, problem analysis, specification

Short Answer Questions

1.	What is an Agile Process / Agile process model?	2× Dec'23, Aug'22
2.	Illustrate any two agile process models.	May'25
3.	Why do ASD project teams have a higher likelihood of success? Explain.	Jun'23
4.	What is the significance of Scrum?	Nov'25
5.	What is the Dynamic Systems Development Method (DSDM)?	Jun'24
6.	Mention the main phases in the requirements engineering process.	Dec'21
7.	Define functional and non-functional requirements.	Jun'22
8.	Identify the non-functional requirement issues considered for any given problem description.	Jan'23
9.	What happens when requirement validation uncovers an error? Who is involved in correcting it?	Dec'24

Long Answer Questions

Agility & Agile Process Models

1.	Provide an overview of Agile development methodologies focusing on XP. What are the key principles and practices of XP? / Explain the XP process model.	2× Jun'24, Aug'23
2.	List the important activities of XP. Explain in detail.	Jun'23
3.	What is a spike solution in XP? Describe the XP concepts of refactoring and pair programming in your own words.	2× Jan'23, Nov'25
4.	Why does an iterative process make it easier to manage change? Is every agile process iterative? Explain using XP.	Dec'24
5.	Explain Scrum and XP methodologies.	May'25
6.	Compare and contrast any two agile process models.	May'25
7.	Illustrate Adaptive Software Development (ASD) in the agile process model.	Dec'23
8.	Explain the features and applications of the Feature Driven Development (FDD) agile model.	Nov'25
9.	Most agile models recommend face-to-face communication, yet teams and customers may be geographically separated. Is separation something to avoid? How can it be overcome?	Dec'24
10.	What is Agile Project Management and how did it revolutionise the way software products are built?	Jun'23
11.	Explain why rapid delivery and deployment of new systems is often more important to businesses than detailed functionality.	Jan'23

Requirements Engineering

12.	Explain all the steps in the requirements engineering process / What is the purpose of RE? Describe any two of: elicitation, elaboration, validation.	**2×** Jun'24, Aug'23
13.	Explain elicitation techniques / Discuss any one effective technique for requirements elicitation / Describe the steps in requirement elicitation.	**3×** Jun'23, Jun'24, May'25
14.	What is requirements elicitation? Explain the activities performed, with a watch system (set time and alarm) as an example.	Jun'22
15.	List and explain the Collaborative Requirements Gathering guidelines.	Dec'23
16.	Develop a requirements gathering “kit”: guidelines for a gathering meeting and materials to help create lists and define requirements.	Nov'25
17.	Discuss problems that occur when requirements must be elicited from three or four different customers.	Dec'24
18.	Why is it important to distinguish between developing user requirements and system requirements?	Nov'25
19.	Illustrate validating requirements.	Dec'23
20.	How does analysis modeling help capture unambiguous and consistent requirements? Discuss several methods for requirements validation.	Jun'22
21.	Who should be involved in a requirements review? Draw a process model showing how a review might be organized.	Dec'24

SRS & Requirements Specification

22.	What is SRS? Describe its format, contents/components (with an example template) and importance.	**4×** Jun'22, Dec'21, Jun'23, May'25
23.	What is the structure of a requirements document?	Jun'24
24.	What are the desirable characteristics of a good SRS?	Jan'23
25.	Identify and explain the non-functional requirements of an SRS.	Dec'23
26.	List the functional and non-functional requirements for an online shopping system.	Aug'23
27.	Prepare a software requirements specification for a railway reservation system.	Aug'22
28.	Explain the different specification languages used in requirement specification.	Jun'24
29.	For a conference management site (authors submit papers, chairs assign reviewers and select papers, reviewers enter reviews), list the major use cases with goals, preconditions and exception scenarios.	Jan'23

UNIT III — Project Planning & Estimation

Effort estimation, scheduling & staffing, quality planning, risk management; decomposition — sizing, problem-based, LOC- & FP-based estimation, COCOMO

Short Answer Questions

1.	What is the role of effort estimation in a project, and why is it important to do it early?	Jan'23
2.	What is meant by activity and milestone? Identify three milestones for the house-construction phase.	Jun'23
3.	What is Problem-Based Estimation? / List the problem-based estimation guidelines.	3× Nov'25, Dec'24, Dec'23
4.	What is the role of lines of code in project size estimation?	Aug'22
5.	What is the COCOMO model in software project estimation?	May'25
6.	Explain Risk Management.	Jun'24
7.	Mention the two characteristics of risks.	Dec'21
8.	Compare/contrast reactive and proactive risk strategies.	2× Dec'21, Aug'23
9.	How can various types of risks impact project management processes?	Jun'24

Long Answer Questions

Project Plan, Scheduling & Staffing

1.	Briefly explain the purpose of each section of a software project plan.	Dec'24
2.	What activities should be present in a project plan? Does your answer depend on the SDLC used?	Dec'24
3.	Explain why project planning is iterative and why a plan must be continually reviewed.	Jan'23
4.	Given a detailed schedule with milestone estimates from the project plan, how will you check it is consistent with the overall plan? What if it is not?	Jan'23
5.	Explain the phases of project scheduling and staffing / different strategies for staffing a software project.	2× May'25, Dec'23
6.	How can you say performance is an important consideration during planning? Justify.	Nov'25
7.	How would the project's progress be affected? Explain.	Jun'23

Estimation Techniques & Sizing

8.	Describe the various software sizing methods.	Nov'25
9.	Explain expert judgment and recursive estimation techniques.	Jun'24
10.	Illustrate problem-based estimation with an example.	May'25
11.	Explain cost estimation using LOC and FP approaches with an example / Distinguish LOC-based and FP-based estimation.	2× Jun'23, Dec'23
12.	Cost estimates are inherently risky. Suggest four ways in which the risk in a cost estimate can be reduced.	Dec'24
13.	Explain size-oriented metrics with examples; justify how they act as indicators of project progress.	2× Dec'21, Aug'23
14.	Give examples of product size metrics and how they aid estimating effort, tracking progress and detecting issues.	Jun'24

15.	What is the LOC metric and how does it measure the size and complexity of software?	Jun'24
16.	Define function point. Explain the procedure to calculate FP with an example / function-oriented metrics / applications of function points.	**3×** Dec'21, Aug'22, Aug'23

Numericals

17.	**LOC-based:** CAD application — 3DGA 6800, UI & control 2300, 2DGA 5300, DB mgmt 3350, graphics display 4950, peripheral control 2100, design analysis 8400 LOC. Productivity 620 LOC/pm, labour rate Rs 8000/month. Compute cost per LOC, effort (person-months) and total cost.	Jan'23
18.	**LOC-based:** Household-robot lawn mower — do a functional decomposition, estimate each function's LOC; with 450 LOC/pm and $7000 per person-month, estimate effort and cost.	Nov'25
19.	**FP:** Compute FP with average weighting factors and CAF — Inputs 50, Outputs 40, Inquiries 35, Files 6, External Interfaces 4.	May'25
20.	**Basic COCOMO:** Project of 400 KLOC — compute effort and development time for organic, semidetached and embedded modes.	Jun'24
21.	**COCOMO:** 200 KLOC project, team with average experience, schedule not tight — compute effort, development time, average staff size and productivity.	May'25
22.	**COCOMO II (application composition):** ATM with 12 screens, 10 reports, ~80 components; average complexity and developer/environment maturity — estimate effort using object points.	Dec'24

Risk Management

23.	What are the steps involved in risk management?	Dec'23
24.	Explain Risk Monitoring and Control.	**2×** Jun'23, Jun'24
25.	List and define categories/types of software risks; explain strategies for risk identification and risk analysis.	**3×** Dec'21, Aug'22, Aug'23
26.	Explain budget risks, project risks and technical risks with examples.	Jun'23
27.	Explain risk identification using a checklist.	Aug'23
28.	Describe the difference between risk components and risk drivers.	Nov'25
29.	Differentiate reactive and proactive risk strategies; how do they contribute to project success?	**2×** Aug'22, Jun'24
30.	Explain / elaborate the RMMM (Risk Mitigation, Monitoring and Management) plan with an example.	**3×** Dec'21, Aug'22, Jun'24
31.	Identify major risks of payroll projects and rank them / As risk analyst for computerizing payroll, discuss the RMMM plan.	**2×** Jun'22, Jun'23

UNIT IV — Design, Architecture & Coding

Function- & flow-oriented modelling (DFDs), coupling, cohesion, architecture & taxonomy of styles, component-level design — principles & guidelines, coding principles, incremental code development, code inspection

Short Answer Questions

1.	What is meant by coupling and cohesion?	Jun'22
2.	What do you mean by good design?	Jun'24
3.	What is the motivation for applying the basic design principles? List them.	Jan'23
4.	Define the Open-Closed principle with a diagram.	Dec'23
5.	What is the relationship between architecture and module-level design?	Dec'24
6.	Mention two coding principles used in software design.	May'25
7.	What is Code Inspection?	Jun'23

Long Answer Questions

Function- & Flow-Oriented Modelling

1.	Explain function-oriented design / What is its goal? Explain the use of structure-chart notation with an example.	2× Jun'24, Jan'23
2.	Explain the principles of structured design methodology.	May'25
3.	What is flow-oriented modelling? Explain with an example.	2× Dec'23, Jun'24
4.	Why is data-flow oriented modelling widely used in requirement analysis? Construct a DFD with a suitable example.	Jan'23

Design Concepts — Coupling, Cohesion, Modularity

5.	Explain the design concepts of cohesion and coupling / short notes on coupling and cohesion.	2× Jun'23, May'25
6.	Identify the levels of cohesion and their implications for code comprehension and reusability.	Dec'23
7.	How are coupling and software portability related? Give examples.	Jan'23
8.	If a global variable x must be used to share data between modules A and B, how would you design their interfaces to minimize coupling? Give an example.	Dec'24
9.	Is it reasonable to say problem domain components should never exhibit external coupling? Which components would?	Nov'25
10.	What is Modular Design? Why is it needed? Explain with an example.	Nov'25

Software Architecture

11.	What is software architectural design? Explain with a block diagram / explain various architectural styles with examples.	4× Jun'22, Jun'23, Jun'24, May'25
12.	List and explain the Taxonomy of Architectural Styles.	2× Dec'23, Jan'23
13.	Illustrate the Data-centered Architecture with a neat sketch.	Jun'22
14.	Explain why you may have to design the system architecture before the requirements specification is complete.	Dec'24

Component-Level Design

15.	Define the design process. List / explain the basic principles of software design.	**3×** Jun'22, Jun'24, May'25
16.	What are the characteristics of a good design?	Jun'24
17.	Explain / illustrate component-level design guidelines.	**2×** Dec'23, Nov'25
18.	Examine the importance of appropriate data structures and algorithms when designing software components.	Dec'23

Coding & Code Inspection

19.	Describe / list the coding principles and guidelines for development of code.	**2×** Jun'23, Dec'24
20.	What is the use of a group review meeting? Outline its guidelines.	Jun'23
21.	Provide the rules for evaluating the summary report of a code review.	Dec'24
22.	**Numerical:** Product of 0.350 KLOC with 9 minor and 5 major defects — compute defect density (per KLOC) for both.	Jan'23

UNIT V — Testing, Reviews, Quality & Deployment

Testing concepts & process, strategies, strategic issues, conventional/validation/system testing, white box, black box, automatic vs manual; informal reviews & FTRs; quality concepts, McCall's & HP attributes; deployment planning & rollback

Short Answer Questions

1.	What is Testing? State the objectives of software testing.	2× Jun'24, Dec'21
2.	Explain the difference between an error and a defect.	Jan'23
3.	List out the levels of testing / the software testing steps.	2× Jun'22, Aug'23
4.	Illustrate how test coverage is helpful in measuring the effectiveness of testing.	Aug'22
5.	Is unit testing possible or even desirable in all circumstances? Justify with examples.	Dec'24
6.	What is System testing? Why is it important? Give an example.	2× Dec'23, Jun'23
7.	What does Validation represent?	Jun'22
8.	Who should perform the validation test — developer or user? Justify.	Nov'25
9.	Distinguish alpha testing and beta testing.	Dec'21
10.	What is black box testing and how is it conducted without accessing internal structure?	Jun'24
11.	Differentiate Black-Box and White-Box testing strategies.	Nov'25
12.	Define software quality.	2× Aug'23, Aug'22
13.	List McCall's quality factors / any two software quality attributes.	2× Jun'23, May'25

Long Answer Questions

Testing Concepts & Strategies

1.	Explain why testing can only detect the presence of errors, not their absence.	Dec'24
2.	Describe the characteristics of software testing.	Dec'21
3.	Describe different software testing strategies / list the strategic steps used for testing.	2× May'25, Aug'23
4.	Explain the various strategic issues of testing.	Jun'23
5.	What are the levels of software testing and how do they differ in scope, objectives and techniques?	Jun'24
6.	What is unit testing? Why is it important, how does it improve quality, and what tools are used?	2× Jun'22, Jun'23
7.	Explain System testing and debugging.	Jun'24
8.	Identify the importance of test case design and the factors in designing effective test cases.	Dec'23

Verification & Validation Testing

9.	Explain software verification and validation / the difference between them.	2× Jun'23, Jun'24
10.	Describe verification vs validation. Do both use test-case design methods and testing strategies?	Jan'23
11.	Outline the testing strategies under Validation Testing / how validation testing differs from verification and what techniques it uses.	2× Jun'23, Jun'24
12.	Compare / differentiate alpha testing and beta testing.	2× Aug'23, Aug'22

13.	What are the benefits of involving users in release testing early? Are there disadvantages?	Dec'24
	White Box Testing	
14.	What is white box testing? Explain its methods with an example system / write a detailed note on it.	**4×** Jun'22, Dec'23, Dec'24, Aug'22
15.	Discuss basis path testing with examples; identify the steps to derive test cases.	**2×** Dec'21, Jun'23
16.	What is cyclomatic complexity? Ways to compute it, with an example program; steps to select paths in data flow testing.	**2×** Jun'22, Nov'25
17.	What is the use of flow graph notation? How do you identify the number of independent paths?	Jan'23
18.	**Numerical:** Flow graph with nodes a, (b,c), f, g, h, i, (d,e), j, k and edges a -> bc, bc -> f, bc -> de, f -> g, f -> h, g -> i, h -> i, i -> j, de -> j, j -> a, a -> k. Compute cyclomatic complexity and the number of independent paths.	Jun'23
19.	Explain how the various types of loops are tested.	Nov'25
	Black Box Testing	
20.	Explain black box testing techniques / equivalence partitioning and boundary value analysis.	**3×** Dec'21, Jun'22, Aug'23
21.	Explain boundary value analysis with an example / its use in good test case design.	**3×** Jun'23, Jan'23, Aug'22
22.	Explain boundary conditions and their criteria.	Jun'24
23.	Compare white box and black box testing (approach and coverage) / explain black box and glass box methods with examples.	**3×** May'25, Jun'23, Jun'24
	Software Reviews	
24.	Distinguish between informal reviews and formal technical reviews.	Dec'23
25.	An FTR is effective only if everyone prepares in advance. How do you recognize an unprepared participant? What do you do as review leader?	**2×** Jan'23, Nov'25
	Software Quality	
26.	Explain / describe McCall's quality factors (quality attributes).	**5×** Dec'21, Aug'22, Dec'24, May'25, Nov'25
27.	Explain David Garvin's quality dimensions.	Jun'23
28.	Define software quality and explain its significance in development and user satisfaction / core concepts of software quality.	**2×** Dec'23, Jun'24
29.	What key factors determine software quality, and how can they be measured and improved? / Explain how software quality is achieved.	**2×** Jun'23, Aug'23
30.	Explain quality attributes such as reliability, maintainability, usability, efficiency and portability, and approaches to assess and enhance each.	Jun'23
	Deployment	
31.	Explain the process of deployment and rollback planning.	May'25